

ABIRAMI MUTHIAH

Data Scientist & AI Engineer

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[LinkedIn](#) | [Github](#) | [Portfolio](#)

PROFESSIONAL OBJECTIVE

Data Science final-year student specializing in AI/ML systems, with hands-on experience building RAG pipelines, Computer Vision models, and production-ready deployments on AWS. Seeking a full-time Data Scientist or AI Engineer role from July 2026 where I can drive impactful, scalable AI solutions.

EDUCATION

Multimedia University, Cyberjaya, Malaysia

Bachelor, Computer Science (Hons.) (Data Science)

Aug 2023 - Jul 2026

Multimedia University, Cyberjaya, Malaysia

Foundation in Engineering - CGPA: 3.72

Jul 2022 - Jul 2023

Dean's List Trimester 3- 2021/2022, Dean's List Trimester 1- 2022/2023

PROJECTS

Wafer Defect Detection — Wafer Defect Detection using AI | [Github](#)

- Engineered a YOLO-based Computer Vision pipeline for semiconductor wafer defect localization, achieving 95% detection accuracy with 450ms inference per wafer — directly applicable to production line quality inspection and manufacturing yield improvement
- Automated visual inspection processes, reducing defect detection time from minutes to milliseconds per wafer compared to manual review

E-Invoicing — AI-Powered E-Invoice Application | [Github](#)

- Built a Flask-based e-invoicing system with SQL backend featuring automated invoice generation, validation, and management
- Integrated AI-assisted logic reducing data entry errors by 30% and processing invoices 5x faster than manual workflows
- Designed RESTful API architecture with real-time data handling and operational analytics dashboard

Sentilytics — AI Sentiment Analyzer | [Github](#)

- Developed a multilingual sentiment analysis platform (Sentilytics) using VADER and Multilingual BERT to classify YouTube and social media comments across English, Tamil, and Malay
- BERT outperformed VADER by 15% F1-score, demonstrating superior multilingual contextual understanding
- Built an interactive Streamlit dashboard for real-time sentiment visualization and comment analysis
- Applied feature engineering, hyperparameter tuning, and cross-validation to optimise model performance

AI Humaniser — NLP-Based Text Humanisation Platform | [Github](#)

- Built a full-stack AI platform using FastAPI and Next.js to transform AI-generated text into natural, human-like writing using Transformers, spaCy, and NLTK
- Engineered a multi-stage NLP pipeline covering semantic rewriting, sentence restructuring, tone optimization, and grammar refinement
- Designed a subscription-based SaaS architecture with Free/Plus/Pro tiers, daily quota limits, and real-time text transformation via REST API

COURSES & TRAINING

- Developing Applications in Python on AWS , AWS
- Create Machine Learning Models in Microsoft Azure , Microsoft
- Microsoft Azure Machine Learning for Data Scientists , Microsoft
- Python for Data Science, AI & Development , IBM

EXPERIENCE

Data Science & AI Intern (Cloud Basha Sdn Bhd) (4 months)

- Engineered and deployed the E-Invoice system for client workflows at Cloud Basha, reducing manual data entry workload by ~40% through AI-driven automation
- Designed and deployed a Generative-AI solution (RAG pipeline) for a domain-specific AI platform, improving context-aware document retrieval accuracy for professional workflows
- Developed Analytics Systems backend using REST APIs, MongoDB Atlas, and Redis to support real-time data processing
- Deployed production-ready applications on AWS EC2, managing end-to-end deployment, system documentation, and stakeholder presentations

Clinic Admin & Marketing Assistant (Part-Time) - Klinik Pergigian i-Dental

- Managed and maintained structured digital patient records with data tracking and validation using Excel
- Supported digital operations and system-based workflows in a healthcare environment

SKILLS

- Programming: Python, SQL
- Data & Analytics: Power BI, Excel, Pandas, NumPy, PyTorch, TensorFlow, OpenCV
- Databases & Tools: MongoDB Atlas, Redis, Flask, FastAPI, Streamlit, Postman, Git, GitHub, Firebase
- Cloud & DevOps: AWS (EC2), Microsoft Azure, Docker
- Machine Learning & AI:
 - GenAI & LLM: LLM, Generative AI, RAG, Prompt Engineering, LangChain, n8n AI Workflow, MCP Server, Agentic AI, LLMOps
 - ML & CV: YOLO (Computer Vision), BERT, NLP, HuggingFace Transformers, scikit-learn, MLOps, MLflow, Model Deployment, Model Evaluation
 - NLP Tools: spaCy, NLTK, VADER